Quirk Table I reproduction

arXiv:1708.02243 · colored m_Q = 1.8 TeV, Λ = 2 keV · EW m_Q = 800 GeV, Λ = 4 keV · results as of 5 October 2026 · Wilson 95% intervals · p = exact binomial against the paper's value

Trigger and fiducial efficiency (the same in every SCT setting)

Benchmark	Sample	k / n	ε [Wilson 95%]	Paper	p	Verdict
ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching) Same cut as the paper (Sec. III B), on the final-state quirk pair.						
Colored	COL1000	101 / 419	0.241 [0.203, 0.284]	0.24	0.95	**No statistically significant inconsistency**
	COL2000	207 / 873	0.237 [0.210, 0.266]	0.24	0.87	**No statistically significant inconsistency**
EW	EW1000	89 / 511	0.174 [0.144, 0.209]	0.10	2.8×10^{-7}	**Strongly inconsistent**
	EW2000	186 / 1045	0.178 [0.156, 0.202]	0.10	1.8×10^{-14}	**Strongly inconsistent**
ε_{fid} = N(both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered) The paper's definition (Sec. IV); colored: both R-hadrons charged, in numerator and denominator.						
Colored	COL1000	6 / 27	0.222 [0.106, 0.408]	0.28	0.67	**No statistically significant inconsistency**
	COL2000	22 / 68	0.324 [0.224, 0.442]	0.28	0.42	**No statistically significant inconsistency**
EW	EW1000	33 / 89	0.371 [0.278, 0.475]	0.28	0.059	**Borderline / mild tension**
	EW2000	62 / 186	0.333 [0.270, 0.404]	0.28	0.12	**No statistically significant inconsistency**

Four ways to set the SCT resolution along the beam (z)

	SCT z: signal / background	ε_{reco} COL2000 (paper 0.35)	ε_{reco} EW2000 (paper 0.72)	Conversion + pile-up in signal region (paper ≈ 2 of 9139)	Total ε COL / EW (paper 0.024 / 0.020)	Signal / background like the paper?
A	no smearing / 580 μm	8/22 = 0.36 [0.20, 0.57] p = 1.00	50/62 = 0.81 [0.69, 0.89] p = 0.16	0 / 9139 (UL 3.3×10^{-4}) cited material model	0.028 / 0.048	**signal ✓ background ✓**
B	580 μm / 580 μm	3/22 = 0.14 [0.05, 0.33] p = 0.042	14/62 = 0.23 [0.14, 0.34] p = 9.6×10^{-16}	0 / 9139 (UL 3.3×10^{-4}) cited material model	0.010 / 0.013	**signal ✗ background ✓**
C	1.16 mm / 1.16 mm	1/22 = 0.05 [0.01, 0.22] p = 0.0013	10/62 = 0.16 [0.09, 0.27] p = 7.8×10^{-20}	0 / 9139 (UL 3.3×10^{-4}) no material model	0.003 / 0.010	**signal ✗ background ✓**
D	no smearing / no smearing	8/22 = 0.36 [0.20, 0.57] p = 1.00	50/62 = 0.81 [0.69, 0.89] p = 0.16	146 / 9139 = 1.6% cited material model	0.028 / 0.048	**signal ✓ background ✗**

Background: converted photon + μ = 50 pile-up, 9139 photons, Ref. [27] lengths, fit from the vertex. Pile-up only (5000 events, 1.16 mm): 0 in the signal region. Total ε uses the 2000-event samples. ✓ = paper value inside our Wilson 95% interval (signal: p ≥ 0.05 for both samples).

Reconstruction efficiency for every SCT setting

Sample	SCT z (signal)	k / n	ε [Wilson 95%]	Paper	p	Verdict
ε_{reco} = N(fiducial, plane found, 2 hits in 7 layers + ≥ 1 in the 8th, Δw < 1 cm, Δs < 0.01 cm) / N(fiducial) The paper's plane algorithm and signal-region cuts (Sec. III C, IV).						
COL1000	no smearing	5 / 6	0.833 [0.436, 0.970]	0.35	0.022	**Statistically inconsistent**
	580 μm	2 / 6	0.333 [0.097, 0.700]	0.35	1.00	**No statistically significant inconsistency**
	1.16 mm	1 / 6	0.167 [0.030, 0.564]	0.35	0.67	**No statistically significant inconsistency**
COL2000	no smearing	8 / 22	0.364 [0.197, 0.570]	0.35	1.00	**No statistically significant inconsistency**
	580 μm	3 / 22	0.136 [0.047, 0.333]	0.35	0.042	**Statistically inconsistent**
	1.16 mm	1 / 22	0.045 [0.008, 0.218]	0.35	0.0013	**Strongly inconsistent**
EW1000	no smearing	27 / 33	0.818 [0.656, 0.914]	0.72	0.25	**No statistically significant inconsistency**
	580 μm	5 / 33	0.152 [0.067, 0.309]	0.72	1.6×10^{-11}	**Strongly inconsistent**
	1.16 mm	2 / 33	0.061 [0.017, 0.196]	0.72	2.0×10^{-15}	**Strongly inconsistent**
EW2000	no smearing	50 / 62	0.806 [0.691, 0.886]	0.72	0.16	**No statistically significant inconsistency**
	580 μm	14 / 62	0.226 [0.140, 0.344]	0.72	9.6×10^{-16}	**Strongly inconsistent**
	1.16 mm	10 / 62	0.161 [0.090, 0.272]	0.72	7.8×10^{-20}	**Strongly inconsistent**

What the paper and its references say

Paper, Sec. III A (signal detector)
“We hereby use the parameters of the various detector elements as specified in [27]. Specifically, for the pixel detector we use the pixel size rather than the resolution and for the silicon microstrip tracker (SCT) we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27].”

Ref. [27] (ATL-PHYS-PUB-2009-002, in arXiv:0901.0512, p. 18 = PDF p. 42)
“The intrinsic accuracies per module in the barrel are 17 μm (R−φ) and 580 μm (z)”

Paper, Sec. III D (backgrounds)
“We model pile-up by randomly selecting minimum bias events ... processed by the simplified detector described in [34]. ... The e^+e^- pair is then passed through the same detector simulation as described for the pile-up background.”

Ref. [34] (arXiv:1612.00850), App. A
“We take the resolution and geometry of each component from the ATLAS technical design report”

Paper, Sec. IV (signal result)
“the reconstruction efficiency for these cuts can be as high as ∼73%.”

Paper, Sec. IV (background result)
“Even though Fig. 5 suggests a few background events after the final selection cut of Δs < 0.01 cm, we suspect that they can easily be removed”

Paper, Sec. III A (acceptance)
“For simplicity, we only included the barrel of the pixel and SCT detectors in our simulations, which effectively restricts the fiducial range to $|\eta| \lesssim 1.8$.”

How each scenario rests on these lines

- **B and C** follow the signal sentence literally: “twice the intrinsic accuracy” of [27] gives 1.16 mm along the beam (C); the undoubled [27] value is 580 μm (B). The paper does not say which direction is doubled.
- **A** follows the paper's results: no smearing reproduces the ≈ 73% and the Fig. 5 signal shape, and the background uses the cited detector [34], whose resolution comes from the ATLAS TDR (≈ 580 μm). No sentence states “no smearing” for the signal.
- **D** (one setting, no smearing) reproduces the signal but gives 146 / 9139 conversions in the signal region, against the paper's “a few”.

Definitions

ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching)
ε_{fid} = N(triggered, both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered)
ε_{reco} = N(fiducial, plane found, 2 hits in 7 layers + ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm) / N(fiducial)
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

Colored: both R-hadrons must be charged; this is applied to the triggered events before ε_{fid} (27 of 101 and 68 of 207 events).

Wilson 95% interval for k of n ($\varepsilon = k/n$, $z = 1.96$): $[\ \varepsilon + z^2/2n \pm z\sqrt{(\ \varepsilon(1-\varepsilon)/n + z^2/4n^2\ )}\ ] / (1 + z^2/n)$
95% upper limit when 0 of n pass: $UL = 1 - 0.05^{1/n} \approx 3/n$
p = exact binomial probability, if the paper's value were exactly right, of a count as unlikely as ours or more

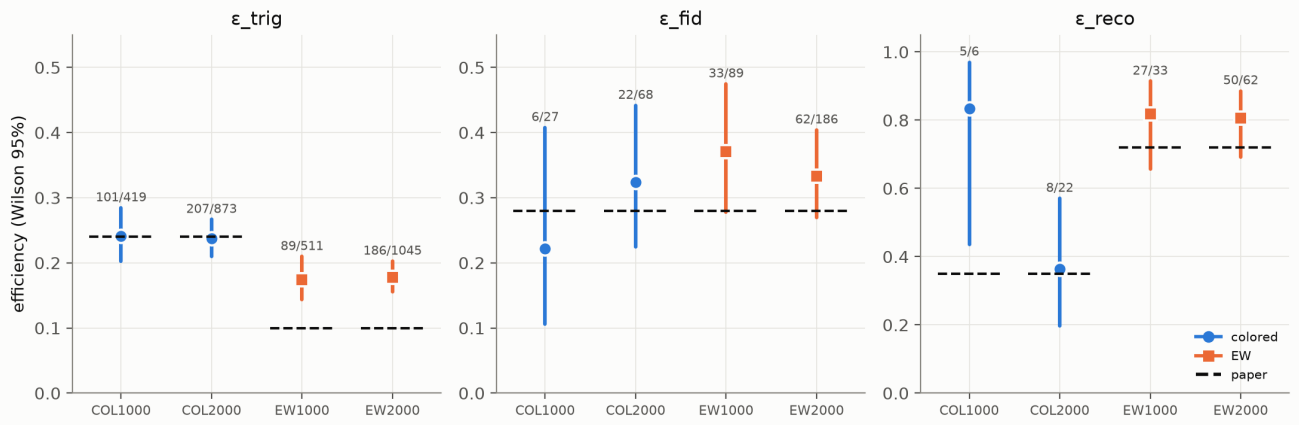
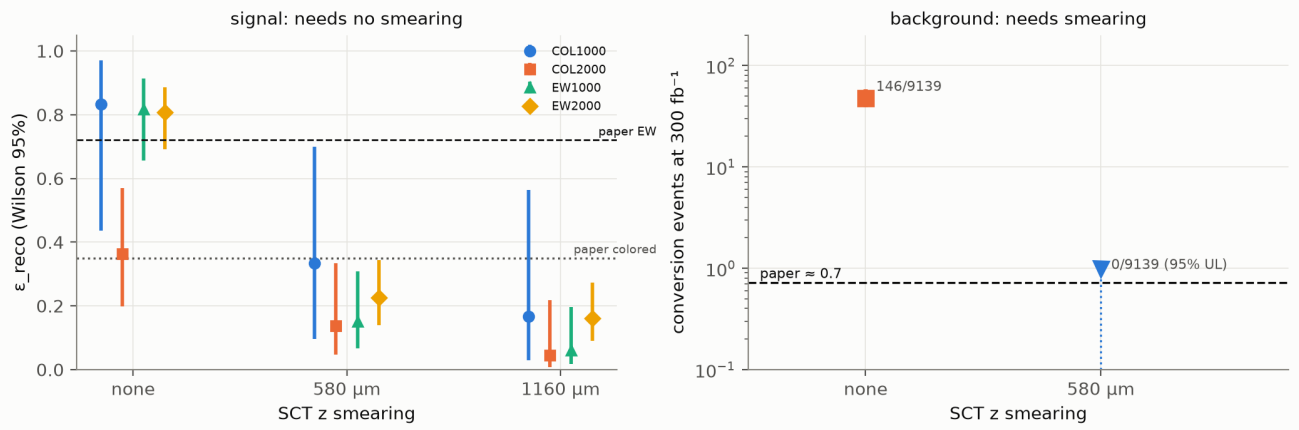
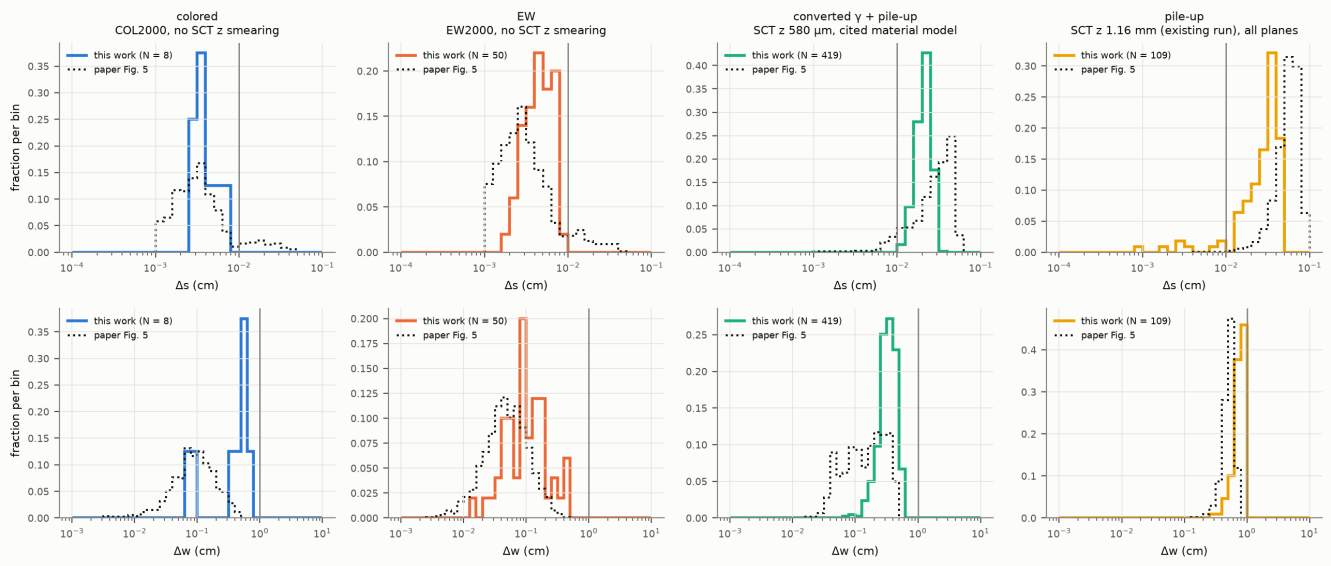


Table I efficiencies, each sample separately; bars = Wilson 95%; dashed = paper. ε_{reco} shown for scenario A (no SCT smearing along the beam).



Left: ε_{reco} for each SCT z setting. Right: converted-photon + pile-up events in the signal region at 300 fb^{-1} (paper σ ≈ 1 pb, 1% conversion); triangle = 95% upper limit for 0 events.

Paper Figure 5: Δs and Δw



Δs (top) and Δw (bottom), unit area on the paper's bins; dotted = paper's Fig. 5. Scenario A. Signal: no SCT smearing along the beam. Converted γ + pile-up: SCT z 580 μm with the cited material model. Pile-up: existing run (SCT z 1.16 mm), all planes. Our Δw is $\sqrt{\lambda_2}$ (paper Eq. 11); the paper's plotted Δw appears to be $\frac{1}{2}\sqrt{\lambda_2}$.

Paper Figures 2, 3 and 4

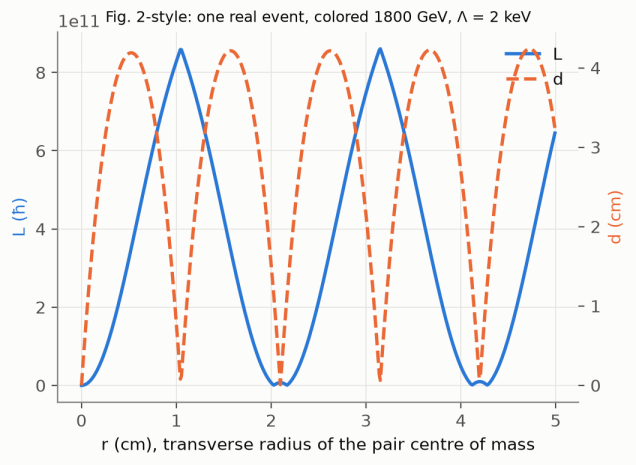


Fig. 2: angular momentum L and quirk separation d for one colored event (COL2000 event 107). L reaches ≈ 9×10¹¹ ħ, the paper's ≈ 10¹² ħ scale.

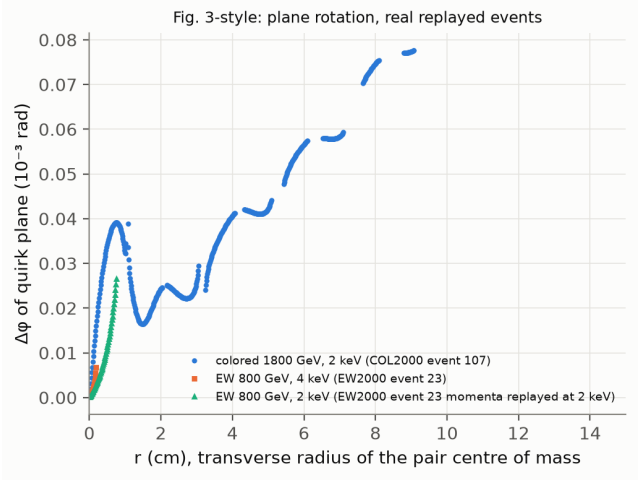


Fig. 3: rotation of the quirk plane stays below 10^{-4} rad, so the hits stay on one plane.

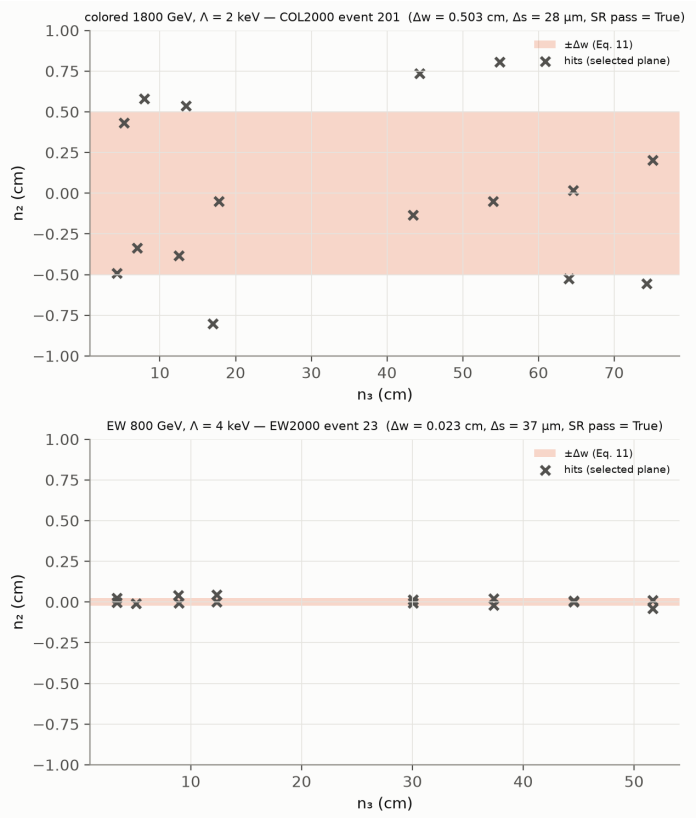


Fig. 4: hits of one colored (COL2000 event 201) and one EW (EW2000 event 23) event in the plane's own coordinates, scenario A (no SCT smearing); both pass the signal region (Δs = 28 and 37 μm). Band = ±Δw.